SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems.* (BL3, BL6)

Activity Tool : Scenario-Based Task (High)
Assessment Tool Weightage: 35%

Dr.G.Ayyappan

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Given the scenario of a School Management System, identify relations and write SQL to list all students with their class teacher and grade.	BL3 – Apply	**5**
2	A hospital wants to track patient appointments. Design the relational schema and write SQL to find doctors with more than 10 appointments this month.	BL3 – Apply	**5**
3	In an e-commerce scenario, write SQL queries using sub-queries to find products that have never been ordered.	BL3 – Apply	**5**
4	For a university database, write relational algebra expressions to find students enrolled in both 'DBMS' and 'OS' courses.	BL3 – Apply	**5**
5	Given a payroll scenario, create a view showing employees whose salary is below department average and write a query to update their salary by 10%.	BL6 – Create	**5**
6	In a banking scenario, apply JOIN operations to retrieve customer account details, transaction history, and branch information.	BL3 – Apply	**5**

7	A retail store needs daily sales summary. Write SQL with GROUP BY and HAVING to report total sales per category exceeding Rs. 10,000.	BL3 – Apply	**5**
8	For a library system scenario, construct tuple relational calculus expressions to find members who borrowed books from all genres.	BL3 – Apply	**5**
9	Given an inventory management scenario, define CHECK and FOREIGN KEY constraints and demonstrate their enforcement through SQL.	BL3 – Apply	**5**
10	Design a relational schema for an HR system and create materialized views for monthly attendance and performance summaries.	BL6 – Create	**5**

Code of Conduct:

I Hemalakshmi H - 192571049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Scenario based assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

		address the scenario			
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Question 1:

School Management System

Relations Identified

- STUDENT(StudentID, Name, ClassID, Grade)
- CLASS(ClassID, ClassName, TeacherID)
- TEACHER(TeacherID, TeacherName)

SQL Query

```
SELECT S.Name AS StudentName,
       T.TeacherName AS ClassTeacher,
       S.Grade
FROM STUDENT S
JOIN CLASS C ON S.ClassID = C.ClassID
JOIN TEACHER T ON C.TeacherID = T.TeacherID;
```

Question 2:

Hospital Appointment System

Relational Schema

- PATIENT(PatientID, Name, Age, Gender)
- DOCTOR(DoctorID, DoctorName, Specialization)
- APPOINTMENT(AppointmentID, PatientID, DoctorID, Date, Time)

SQL Query

```
SELECT D.DoctorName, COUNT(A.AppointmentID) AS TotalAppointments
FROM DOCTOR D
JOIN APPOINTMENT A ON D.DoctorID = A.DoctorID
WHERE MONTH(A.Date) = MONTH(CURDATE())
GROUP BY D.DoctorID
HAVING COUNT(A.AppointmentID) > 10;
```

Question 3:

E-Commerce Scenario
Find products never ordered

```
SELECT ProductName
FROM PRODUCT
WHERE ProductID NOT IN (
        SELECT ProductID FROM ORDER_ITEM
);
```

Question 4:

University Database

Relational Algebra: Students enrolled in both DBMS and OS

Let DBMS = C1, OS = C2

$$\pi_{Name}((\sigma_{CourseID='C1'}(Enrollment) \bowtie Student) \cap (\sigma_{CourseID='C2'}(Enrollment) \bowtie Student))$$

Question 5:

Payroll Scenario

View for employees below department average

```
CREATE VIEW LowSalaryView AS
SELECT EmpID, EmpName, DeptID, Salary
FROM EMPLOYEE
WHERE Salary < (
         SELECT AVG(Salary)
        FROM EMPLOYEE
        WHERE DeptID = EMPLOYEE.DeptID
);
```

Update salary by 10%

```
UPDATE EMPLOYEE
SET Salary = Salary * 1.10
WHERE EmpID IN (SELECT EmpID FROM LowSalaryView);
```

Question 6:

Banking Scenario

JOIN to retrieve customer, account, transactions, branch

```
SELECT C.Name, A.AccountID, A.Balance,
       T.TransactionID, T.Amount, T.Type,
       B.BranchName, B.City
FROM CUSTOMER C
JOIN ACCOUNT A ON C.CustomerID = A.CustomerID
JOIN BRANCH B ON A.BranchID = B.BranchID
LEFT JOIN TRANSACTION T ON A.AccountID = T.AccountID;
```

Question 7:

Retail Store Sales

Daily sales summary > Rs.10,000

```
SELECT P.Category,
       SUM(P.Price * S.Quantity) AS TotalSales
FROM PRODUCT P
JOIN SALES S ON P.ProductID = S.ProductID
WHERE S.SaleDate = CURDATE()
GROUP BY P.Category
HAVING SUM(P.Price * S.Quantity) > 10000;
```

Question 8:

Library System

TRC: Members who borrowed books from all genres

$$\{m.Name \mid MEMBER(m) \land \forall g\ (\exists b(BOOK(b) \land b.Genre = g) \rightarrow \exists br(BORROW(br) \land br.MemberID = m.MemberID \land br.BookID = b.BookID))\}$$

Meaning: For every genre, the member must have borrowed at least one book.

Question 9:

Inventory Management

CHECK and FOREIGN KEY constraints

```
CREATE TABLE ITEM (
    ItemID INT PRIMARY KEY,
    ItemName VARCHAR(50) NOT NULL,
    Price DECIMAL(10,2) CHECK (Price > 0)
);

CREATE TABLE WAREHOUSE (
    WarehouseID INT PRIMARY KEY,
     Location VARCHAR(50)
);

CREATE TABLE STOCK (
    StockID INT PRIMARY KEY,
    ItemID INT,
    Quantity INT CHECK (Quantity >= 0),
    WarehouseID INT,
    FOREIGN KEY (ItemID) REFERENCES ITEM(ItemID),
    FOREIGN KEY (WarehouseID) REFERENCES
    WAREHOUSE(WarehouseID)
);
```

Question 10:

HR System

Relational Schema

- EMPLOYEE(EmpID, Name, DeptID)
- DEPARTMENT(DeptID, DeptName)
- ATTENDANCE(EmpID, Date, Status)
- PERFORMANCE(EmpID, Month, Score)

Materialized View: Attendance

```
CREATE MATERIALIZED VIEW MonthlyAttendance AS
SELECT EmpID, MONTH(Date) AS Month,
    COUNT(*) AS DaysPresent
FROM ATTENDANCE
```

```sql
WHERE Status = 'Present'
GROUP BY EmpID, MONTH(Date);
```

Materialized View: Performance

```sql
CREATE MATERIALIZED VIEW PerformanceSummary AS
SELECT EmpID, AVG(Score) AS AvgScore
FROM PERFORMANCE
GROUP BY EmpID;
```